

# DIFFIE-HELLMAN

## AIM

To implement the **Diffie-Hellman key exchange algorithm** to securely generate a shared secret key between two users.

## ALGORITHM

1. Start the program.
2. Generate the key pair for **Alice**.
3. Generate the key pair for **Bob**.
4. Exchange their public keys.
5. Compute the shared secret using the private key and received public key.
6. Compare both generated shared secrets.
7. Display the shared secret and verify that both are equal.
8. Stop the program.

## PROGRAM:

```
import java.security.KeyPair;
import java.security.KeyPairGenerator;
import java.util.Arrays;
import javax.crypto.KeyAgreement;

public class Main {

    public static void main(String[] args) throws Exception {

        KeyPairGenerator keyGen = KeyPairGenerator.getInstance("DH");
        keyGen.initialize(2048);

        KeyPair alicePair = keyGen.generateKeyPair();
        KeyPair bobPair = keyGen.generateKeyPair();

        KeyAgreement alice = KeyAgreement.getInstance("DH");
```

```
alice.init(alicePair.getPrivate());
alice.doPhase(bobPair.getPublic(), true);
byte[] aliceSecret = alice.generateSecret();

KeyAgreement bob = KeyAgreement.getInstance("DH");
bob.init(bobPair.getPrivate());
bob.doPhase(alicePair.getPublic(), true);
byte[] bobSecret = bob.generateSecret();

System.out.println("Alice Shared Secret:");
for (byte b : aliceSecret)
    System.out.printf("%02X", b);

System.out.println("\n\nBob Shared Secret:");
for (byte b : bobSecret)
    System.out.printf("%02X", b);

System.out.println("\n\nShared Secret Match: " +
    Arrays.equals(aliceSecret, bobSecret));
}
}
```

```
input
Alice Shared Secret:
90421CA42C766C01B1AD7F47C1F839AF862123B4C23C6D20978B94407AD9A18C59F5E043CF09A6911D03B6C79155305D2E8AD7A7A64F523D04DC1853A508B59A0B8C25F85
3951F77581AA49CA5923E16A518A2FCB6AF4A427BC1FEDA5399AD755A0E86AD9BA851829D5B1EE8E482BC109C07EDD94ED7B0C955F87D37ED37900457EFEA86936B36136C
8599D1EFCDE7069CF170B2B308EAF83DA27ED5C6560D160FC82AF1EA74B2F92B3D5203AF54C18CCD2CA9114FB64A47647C6348E83090BD74B472CE1A5853E791ED5671E8F
B87E929538EB2225B67D349A67CADA30904A65A66E373740617371F711A35B60879D5A1436E1147DB6124242D91AFE000121E

Bob Shared Secret:
90421CA42C766C01B1AD7F47C1F839AF862123B4C23C6D20978B94407AD9A18C59F5E043CF09A6911D03B6C79155305D2E8AD7A7A64F523D04DC1853A508B59A0B8C25F85
3951F77581AA49CA5923E16A518A2FCB6AF4A427BC1FEDA5399AD755A0E86AD9BA851829D5B1EE8E482BC109C07EDD94ED7B0C955F87D37ED37900457EFEA86936B36136C
8599D1EFCDE7069CF170B2B308EAF83DA27ED5C6560D160FC82AF1EA74B2F92B3D5203AF54C18CCD2CA9114FB64A47647C6348E83090BD74B472CE1A5853E791ED5671E8F
B87E929538EB2225B67D349A67CADA30904A65A66E373740617371F711A35B60879D5A1436E1147DB6124242D91AFE000121E

Shared Secret Match: true

...Program finished with exit code 0
Press ENTER to exit console.
```

## RESULT

Thus, the **Diffie-Hellman key exchange algorithm** was implemented successfully, and both users generated the same shared secret key.